LLMs under the hood

Tokens, the context window, temperature, and why a model answers questions it can't know.

60 min | Recorded | Class 6 of 15

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2 slides

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3 slides

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Section 1 of 5

Tokens

What the model actually reads, and what you actually pay for.

01

A model doesn't read words; it reads tokens, and our names are expensive

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A model doesn't read words; it reads tokens, and our names are expensive

Text gets chopped into pieces before the model sees it. Common English words are one piece. Company names, IDs and German are several.

Tokens are also the unit you're billed in, and the unit the context window is measured in.

Invoices from Armadillo Security Services not showing in InvoiceHub.
13 tokens

Bayerische Unfug Maschinen
7 tokens

LOL-SH-4472190
8 tokens

Die Rechnung ist noch nicht angekommen.
9 tokens

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Numbers are approximate. Run tokens.py live and show how LOL-SH-4472190 splits into digits.

It’s predicting the **next token**. That’s the whole trick.

Every answer is one token at a time, each picked from a list of likely next tokens.

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Section 2 of 5

The context window

Everything the model can see at once.

01

The context window is the model’s whole world for one request

02

When it overflows, you get an error or, worse, a quiet cut

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The context window is the model's whole world for one request

Your instructions, the ticket, any documents you paste, the chat so far, and the answer it's writing: all of it has to fit.

Anything outside the window doesn't exist for the model. It doesn't remember yesterday's chat unless you send it again.

tokens, a common size 128,000

of plain text about 300 pages

at most roughly 4,000 of our tickets

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When it overflows, you get an error or, worse, a quiet cut

Error

- The provider refuses the request
- Annoying, but you know
- Fix: send less, or summarise first

Silent truncation

- The start of the text is dropped
- The answer looks confident anyway
- Fix: count tokens before you send

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Section 3 of 5

Temperature and making things up

Why the same question gets different answers, and some of them invented.

01
Temperature sets how adventurous the pick is

02
Ask about something it can't know, and it answers anyway

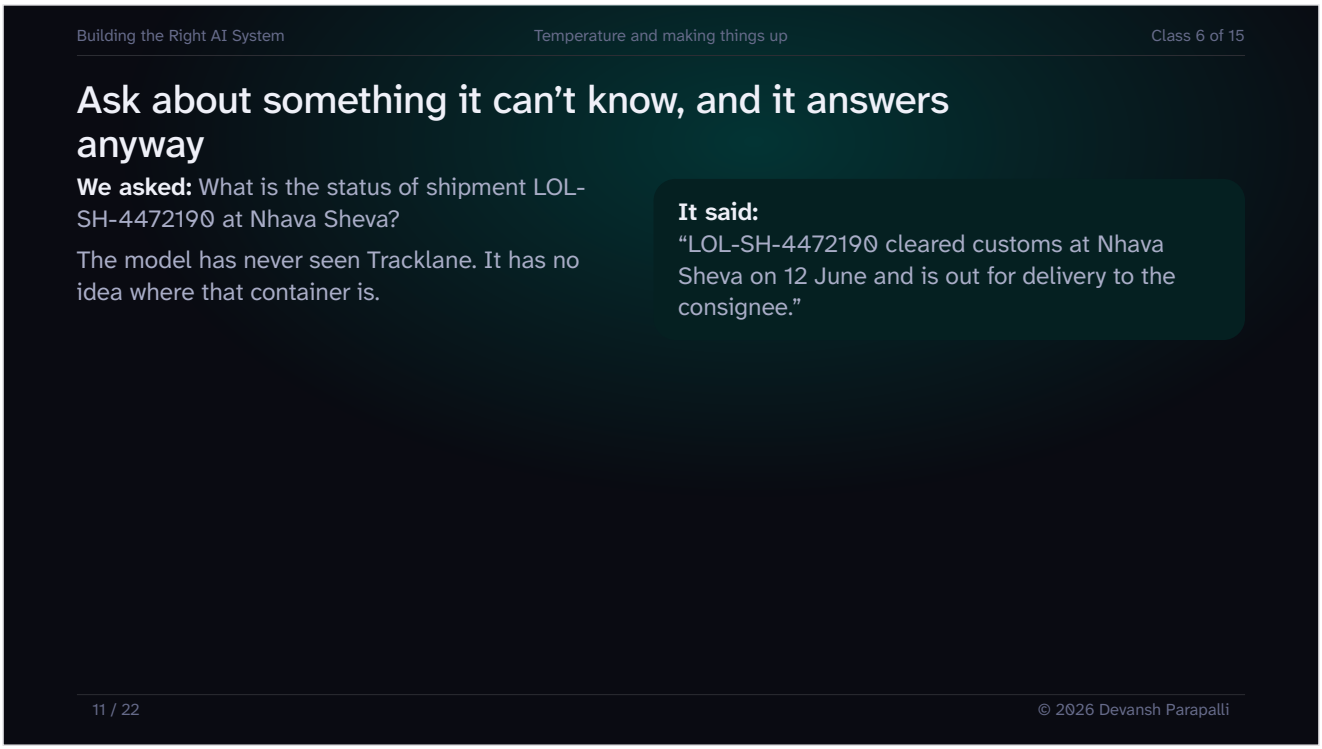
03
Why it makes things up

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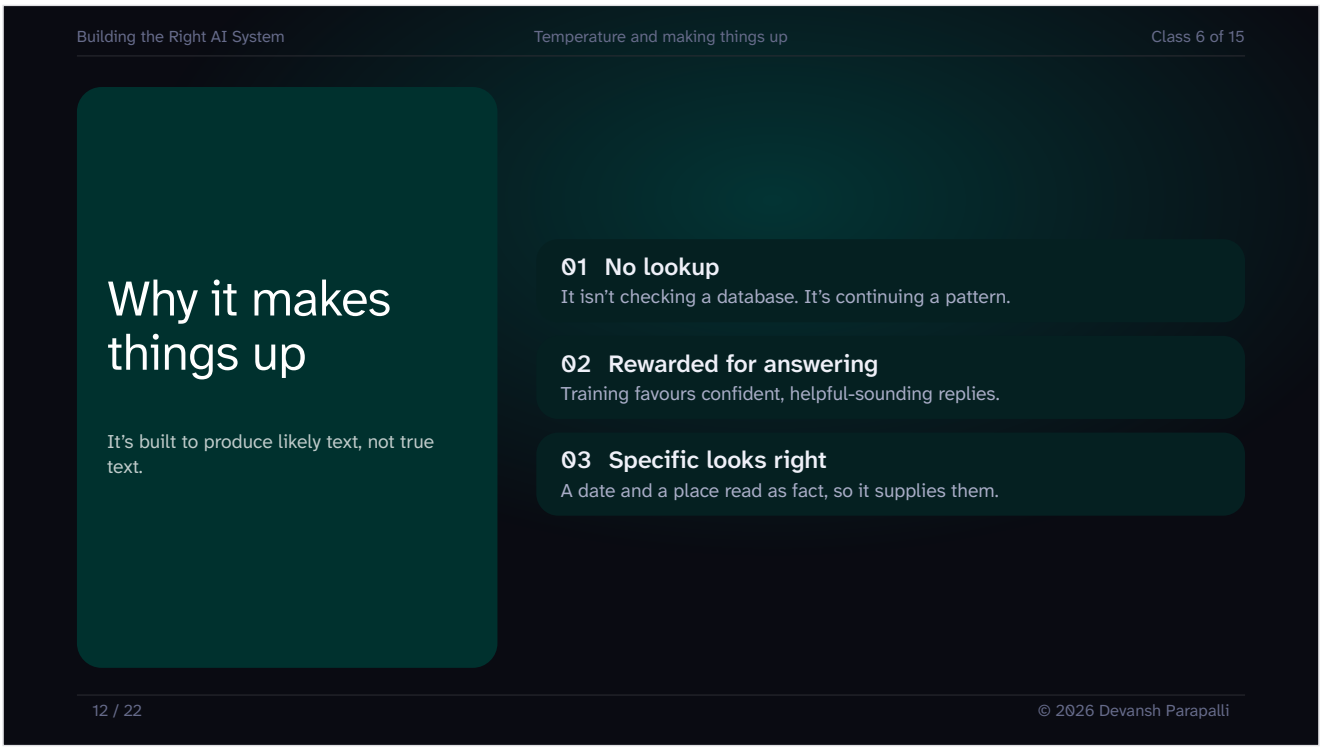
Temperature sets how adventurous the pick is

0	Takes the most likely token nearly every time. Best for routing, extracting, anything a program reads.
0.7	Some variety. Fine for drafting a reply a person will edit.
1.5	Loose. Creative on a good day, nonsense on a bad one. We don't use it at work.

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That’s a hallucination: plausible text with nothing behind it. The fix is never “prompt harder”; it’s giving the model the real data (class 9) or a tool to look it up (class 10).



Section 4 of 5

Who runs the model

Hosted, our gateway, or on a laptop.

01

Hosted models are the strongest; local ones keep data on the machine

02

Running one yourself is a memory problem before it's anything else

03

Routing every ticket with an LLM costs less than people think, and more than a regex

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Hosted models are the strongest; local ones keep data on the machine

Hosted API

- Strongest models, no hardware
- Pay per token
- Our text leaves the building (class 8)

Open model, run locally

- Weights you download and run
- Smaller, a bit weaker
- Data stays on the laptop or server

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Running one yourself is a memory problem before it’s anything else

A model’s size is its number of parameters. Each one needs memory: two bytes at full quality, about half a byte when compressed to 4 bits.

So a laptop with a decent GPU, or a lot of RAM, runs a small model fine. A server runs bigger ones for a whole team.

parameters, a small model	8 billion
memory at 4-bit	about 5 GB
memory at full quality	about 16 GB

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Routing every ticket with an LLM costs less than people think, and more than a regex

300 tickets a day, about 400 tokens in and 150 out each, at $0.50 per million in and $1.50 per million out.

Cheap. Now compare with the classifier from class 3, which costs nothing per call and gets 90%.

illustrative monthly bill	about $4 a month

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Prices vary by provider and change often; the point is the shape of the sum, not the figure.

Section 5 of 5

Code walkthrough

Count tokens, overflow a window, turn the temperature up.

01

Counting tokens takes three lines, so there's no excuse not to

02

Three temperatures, three runs each, and you can see the spread

03

Telling it to admit what it doesn't know helps, but doesn't make it true

04

Where today sits on the card

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Counting tokens takes three lines, so there's no excuse not to

code/class06/tokens.py

```
enc = tiktoken.get_encoding("o200k_base")
for s in SAMPLES:
    ids = enc.encode(s)
    print(f"{len(ids):>3} tokens  {s}")
```


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Three temperatures, three runs each, and you can see the spread

code/class06/temperature.py

```
PROMPT = "Write a one-line subject for a ticket about Rotterdam scanners not syncing."
for t in (0.0, 0.7, 1.5):
    for _ in range(3):
        print(" ", ask(PROMPT, temperature=t).strip())
```


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Telling it to admit what it doesn’t know helps, but doesn’t make it true

code/class06/made_up.py

```
print(ask("What is the current status of LOL shipment LOL-SH-4472190 at Nhava Sheva?"))
print(ask("What is the current status of LOL shipment LOL-SH-4472190 at Nhava Sheva? "
          "If you don't have access to that data, say so and stop."))
```


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Where today sits on the card

1. A script	The steps are known and don't change.
2. A rule	The decision is known and must be explainable line by line.
3. Classical ML	Labelled history, and the input is a table or short text.
4. A small trained model	Lots of text, images or audio, enough labels, cost per call matters.
5. An LLM with a prompt	**Varied language in and out, few labels, someone checks the output.**
6. An LLM plus retrieval, tools or agents	Needs private knowledge, has to act, or takes several steps.
7. Not yet	No ground truth, needs a guarantee, or nobody owns the result.

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What to do next

1. Run `tokens.py` on three of your own ticket titles.
2. Run `made_up.py` and write down whether the second answer was honest.
3. Read the class 7 pre-reading on JSON and schemas.

Questions after class: the course channel.

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